

# SQL Security Log Analysis

Reproducible authentication investigations using SQLite, CSV data, and documented queries

## Dataset

Synthetic authentication events include normal successful logins plus off-hours access, repeated failures, a failure-to-success sequence, multi-user targeting, and unusual geography. The SQLite database and CSV are included.

| Query                  | Analyst question                                               | Key result in sample                                                     |
|------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------|
| Failed logins          | Which attempts failed and from where?                          | 203.0.113.44 generated repeated failures against j.smith.                |
| Off-hours success      | Which successful logins occurred outside normal hours?         | j.smith success at ~02:19 and a.green late-night success require review. |
| Failures by IP         | Which sources target many accounts or generate high volume?    | 203.0.113.44 and 192.0.2.77 stand out for different patterns.            |
| Failures by user       | Which identities are repeatedly targeted?                      | j.smith has concentrated failures.                                       |
| Success after failures | Did guessing activity lead to a successful login?              | j.smith shows >=3 prior failures from the same source before success.    |
| Unusual geography      | Where do rare successful-login geographies appear?             | DE-BE appears once and warrants contextual investigation.                |
| Top source IPs         | Which sources combine failures, successes, and distinct users? | Supports triage prioritization and spray/stuffing analysis.              |
| Incident window        | What happened during 02:00-02:30?                              | Provides a compact incident timeline dataset.                            |

## Files

- sql/security\_log\_queries.sql
- data/security\_events.csv
- data/security\_logs.db
- outputs/sql\_query\_results.txt

## Analyst Guardrails

- SQL identifies patterns; it does not independently prove malicious intent.
- Off-hours activity requires business-hours, timezone, role, VPN, and device context.
- Repeated failures should be correlated with success, MFA, privilege, and post-login activity.
- Use parameterized queries in application code; these portfolio queries are analyst read-only examples.